DSA Problems

Arrays

- Two Sum
- Contains Duplicate
- Best Time to Buy and Sell Stock
- Merge Sorted Array
- Majority Element
- Remove Duplicates from a Sorted Array
- Max Consecutive Ones
- Check if array is sorted and rotated
- Move Zeroes
- Single Number
- Pascal's Triangle
- Rotate Array

(Cont…)

Arrays

- Majority Element II
- Product of Array Except for Self
- Maximum Subarray/ Maximum Sum Subarray/ Kadanes Algorithm
- Maximum Product Subarray
- Container With Most Water
- Missing Number
- Longest Consecutive Sequence
- Set Matrix Zeroes
- Spiral Matrix
- Rotate Image
- Rearrange array elements by sign
- Next Permutation
- Sort Colors

(Cont…)

Arrays

- Subarray sum equals k
- Merge Intervals
- Find the Duplicate Number
- Repeat and Missing Number Array
- Count Inversions
- Search in a 2D Matrix
- Pow(x, n)
- Unique Paths
- 3Sum
- 4Sum
- Largest Subarray with Sum 0
- Subarray with given XOR
- Trapping Rain Water

(Cont…)

Strings

Strings

- Valid Palindrome
- Valid Anagram
- Roman To Integer
- Longest Common Prefix
- Find the index of the first occurrence in a string
- Remove Outermost Parentheses
- Isomorphic Strings
- Largest Odd Number in String
- Rotate String
- Maximum Nesting Depth of the Parentheses
- Sum of beauty of all substrings
- Minimum add to make parentheses valid

(Cont…)

Strings

- Group Anagrams
- Longest Palindromic Substring
- Palindromic Substrings
- Encode and Decode Strings
- Reverse words in a string
- Minimum Characters required to make a string palindromic
- String to Integer Atoi
- Count and Say
- Compare Version Numbers
- Sort Characters by Frequency
- Shortest Palindrome
- Longest Happy Prefix

Backtracking

Backtracking

- Permutations
- Rat in a Maze
- Combination Sum
- Combination Sum II
- Palindrome Partitioning
- Word Search
- N Queens
- Sudoku Solver

Bit Manipulation

Bit Manipulation

- Number of 1 Bits/ Brian Kernighan's Algorithm/ Number of set Bits/ Hamming Weight/ Population Count
- Counting Bits
- Reverse Bits
- Power of Two
- Single Number
- Minimum Bit Flips to Convert Number
- Divide Two Integers
- Subsets
- Sum of Two Integers
- Count Primes
- Pow(x, n)

(Cont…)

Heaps

Heaps

- Kth Largest Element in a Stream
- Kth Largest Element in an array
- Top K Frequent Elements
- Task Scheduler
- Hand of Straights
- Design Twitter
- Maximum Sum Combinations
- Merge K Sorted Lists
- Find Median from Data Stream

LinkedList

LinkedList

- Reverse a Linked List
- Detect Cycle in a Linked List
- Middle of the Linked List
- Merge Two Sorted Lists
- Intersection of two linked lists
- Palindrome Linked List
- Delete the middle node of a linked list
- Odd Even Linked List
- Delete Node in a Linked List
- Linked List Cycle II
- Add Two Numbers

(Cont…)

LinkedList

- Remove Nth Node From End Of List
- Reorder List
- Flattening a Linked List
- Copy List with Random Pointer
- Sort List
- Rotate List
- Reverse Nodes in K Group
- Merge K Sorted Lists

Stack

Stack

- Valid Parentheses
- Implement Stack using Queues
- Next Greater Element I
- Nearest Smaller Element
- Sum of Subarray Minimums
- Next Greater Element II
- Min Stack
- Online Stock Span
- Asterioid Collision
- Sum of subarray ranges

(Cont…)

Stack

- Remove K Digits
- LRU Cache
- Largest Rectangle in Histogram
- Maximal Rectangle
- LFU Cache

Binary Search

Binary Search

- Binary Search
- Search Insert Position
- Kth Missing Positive Number
- Find first and last position of element in sorted array
- Search in a rotated sorted array
- Search in a rotated sorted array II
- Find Minimum in Rotated Sorted Array
- Single Element in a Sorted Array
- Allocate Books
- Aggressive Cows
- Kth Element of Two Sorted Array

(Cont…)

Binary Search

- Longest Increasing Subsequence
- Median of Two Sorted Arrays
- Find Peak Element
- Koko Eating Bananas
- Minimum Number of days to make m bouquets
- Find the smallest divisor given a threshold
- Capacity to ship packages within D Days
- Find a peak element II
- Search in a 2D matrix
- Search in a 2D matrix II
- Split Array Largest Sum

Dynamic Programming

Dynamic Programming

- Climbing Stairs (Count ways to reach the n-th stairs)
- Best Time to Buy and Sell Stock
- Best Time to Buy and Sell Stock II
- Unique Paths
- Unqiue Paths II
- Longest Increasing Subsequence
- 0/1 Knapsack
- Word Break Problem
- House Robber
- House Robber II

(Cont…)

Dynamic Programming

- Decode Ways
- Egg Dropping Puzzle
- Maximum Sum Increasing Subsequence
- Word Break
- Minimum Path Sum
- Coin Change
- Triangle
- Target Sum
- Minimum Failing Path Sum
- Partition Equal Subset Sum
- Edit Distance

(Cont…)

Dynamic Programming

- Delete Operation for Two Strings
- Largest divisible subset
- Number of longest increasing subsequence
- Count Square Submatrices with all ones
- Longest Common Subsequence
- Best time to buy and sell stock with cooldown
- Partition array for maximum sum
- Best time to buy and sell stock with transaction free
- Best time to buy and sell stock III
- Best time to buy and sell stocok IV
- Palindromic Partitioning

(Cont…)

Dynamic Programming

- Matrix Chain Multiplication
- Minimum Cost to Cut a Stick
- Paritition Array into two arrays to minimize sum difference
- Minimum Insertion Steps to make a string palindrome
- Shortest Common Supersequence
- Distinct Subsequence
- Wildcard Matching
- Burst Ballons
- Parsing a boolean expression
- Palindrome Partitioning II
- Maximal Rectangle

(Cont…)

Greedy

Greedy

- N Meetings in One Room
- Lemonade Change
- Assign Cookies
- JUMP Game
- Jump Game II
- Valid Parenthesis String
- Insert Interval
- Merge Intervals
- Non-overlapping intervals
- Candy

(Cont…)

Greedy

- Meeting Rooms 1
- Meeting Rooms 2
- Minimum Platforms
- Job Sequencing Problem
- Candy

Sliding Window

Sliding Window

- Valid Anagram
- Longest Repeating Character Replacement
- Longest substring without repeating characters
- Max Consecutive Ones III
- Binary Subarrays with Sum
- Count Number of Nice Subarrays
- Maximum Points you can obtain from cards
- Number of substrings containing all three characters
- Minimum Window Substring
- Sliding Window Maximum
- Subarrays with K different Integers

Trees

Trees

- Invert/Flip Binary Tree / Mirror Tree
- Inorder Traversal
- Preorder Traversal
- Postorder Traversal
- Count Complete Tree Nodes
- Subtree of Another Tree
- Same Tree
- Symmetric Tree
- Maximum Depth of Binary Tree
- Diameter of Binary Tree

(Cont…)

Trees

- Balanced Binary Tree
- Search in a Binary Search Tree
- Insert into a Binary Search Tree
- Two Sum IV Input is a BST
- Floor From BST
- Ceil From BST
- Left View of Binary Tree
- Bottom View of Binary Tree
- Top View of a Binary Tree
- Right Side View
- Level Order Travesal

(Cont…)

Trees

- Lowest Common Ancestor of a Binary Tree
- Binary Tree Zigzag Level Order Traversal
- Convert Sorted Array to Binary Search Tree
- Delete Node in a BST
- Maximum Width of Binary Tree
- Binary Tree Maximum Path Sum
- Construct Binary Tree from Preorder and Inorder Traversal
- Binary Tree Level Order Traversal
- Validate Binary Search Tree
- Flatten Binary Tree to Linked List
- Populating next right pointers in each node

(Cont…)

Trees

- Kth Largest Element in a BST
- Kth Smallest Element in a BST
- All Nodes Distance K in Binary Tree
- Predecessor and Successor
- Lowest Common Ancestor of a Binary Search Tree
- Recover Binary Search Tree
- Vertical Order Traversal of a Binary Tree
- Serialize and Deserialize Binary Tree

Graph

Graph

- DFS of Graph
- BFS of Graph
- Flood Fill
- Clone Graph Link
- Course Schedule I
- Topological Sort
- Number of Provinces
- Pacific Atlantic Water Flow
- Rotting Oranges
- Number of Islands

(Cont…)

Graph

- Is Graph Bipartite
- Implement Dijkstra Algorithm
- Graph Valid Tree
- Number of Provinces
- Minimum Spanning Tree
- Surrounded Regions
- Number of Enclaves
- Find Eventual Safe States
- 01 Matrix
- Shortest Path in BInary Matrix
- Path with Minimum Effort

(Cont…)

Graph

- Cheapest Flights within K Stops
- Network Delay Time
- Number of Ways to Arrive at destination
- Number of Operations to make network connected
- Most stones removed with same row or columns
- Accounts Merge
- Alien Dictionary
- Word Ladder
- Word Ladder II
- Making a large island
- Swim in rising water

(Cont…)

Trie

Trie

- Implement Trie (Prefix Tree)
- Implement Trie II
- Power Set
- Count Distinct Substrings
- Maximum Xor of Two Numbers in an Array
- Complete String
- Maximum Xor with an element from array

- **HashMap + Doubly LinkedList**